Enhancing Accuracy in Reservoir Drinking Water Quality Assessment Using Machine Learning.

Group 07

S17216 - B.K.S.Basnayake
S17229 - T.K.I.Desaa
S17240 - H.A.P.Ashani
S17244- K.D.S.S.N.Hemachandra
S17245 - S.D.Jayasundara
S17284 - R.P.I.D.Wickramasinghe

Problem Statement

- Traditional water quality classification methods:
  - Fail to capture complex interactions among multiple parameters
  - Focus mainly on overall accuracy
  - Do not prioritize minimizing False Negatives (unsafe water classified as safe)

  This may lead to serious public health risks.

Literature Review & Research Gap

- **Existing Study (JISEM, 2024)**
  Applied Support Vector Machine (SVM) for drinking water classification
  Classified water into Potable and Non-potable categories

- **Model Performance:**
  Overall Accuracy: 64%
  Precision:
  Potable Water → 0.56
  Non-Potable Water → 0.71

- **Identified Research Gap**
  1. Limited analysis of combined feature interaction effects
  2. Primary focus on overall accuracy rather than safety-critical error reduction
  3. Less emphasis on minimizing False Negatives (Unsafe water incorrectly classified as safe)

Objectives

➢ Main Objective

To develop a machine learning-based model for accurate classification of drinking water suitability by analyzing the combined effects of water quality parameters.

➢ Specific Objectives

- To minimize False Negatives to prevent unsafe water from being misclassified as safe.
- To evaluate model performance using Accuracy, Precision, Recall, and F1-score.
- To analyze combined feature interactions.
- To identify influential parameters.

Methodology

Water Quality Dataset

Model Selection

Data Processing (Missing Values Handling, Feature Selection)

Exploratory Data Analysis (Correlation Matrix, Feature Distribution)

Model Training (Train–Test Split (80% / 20%, Random Forest Classifier)

Model Evaluation & Prediction of Water Suitability(Accuracy, Confusion Matrix, Metrics)

Model Selection

Model	Test Accuracy	Comments
Logistic Regression	79.12%	Performs poorly due to linear assumptions; cannot model complex patterns
SVM (RBF)	78.02%	Nonlinear model but sensitive to parameter scaling & dataset imbalance
KNN (k=5)	77.66%	Weak performance due to mixed feature scales & noisy boundaries
Random Forest	96.70%	Best performance; handles nonlinear interactions; robust; low error rate
Cross-Validation (RF)	95.59%	Confirms stability and prevents overfitting

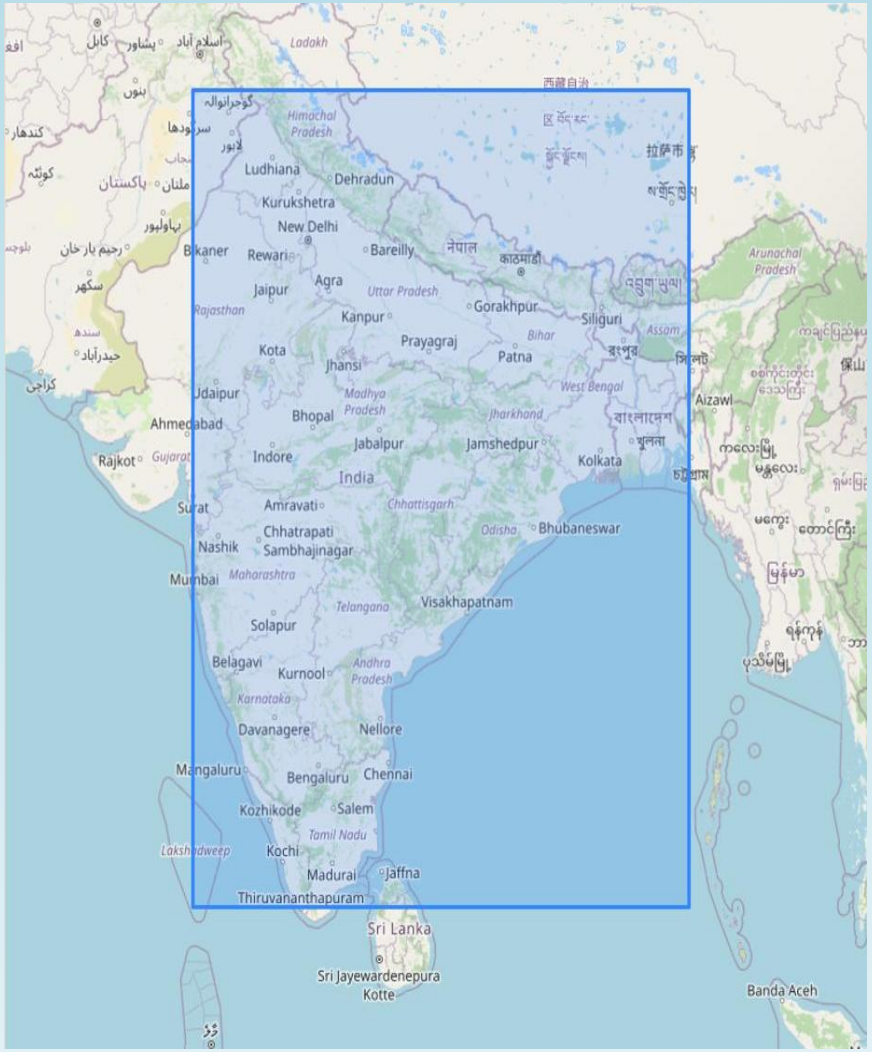
Model : Random Forest Classifier

- Handles nonlinear relationships.
- Provides feature importance analysis.
- Shows highest performance in both test accuracy and cross-validation

Model Training & Validation

- Data splitting - Dataset divided into Training set (80%) & Testing set (20%)
- Model training - Random Forest Classifier trained on training dataset.
- Overfitting avoided by:
  - Testing model performance on unseen data
  - Ensuring no data leakage during preprocessing
- Model validated using multiple evaluation metrics:
  Accuracy, Precision, Recall, F1-score, Confusion Matrix
- Cross-validation suggested for future work to confirm robustness.

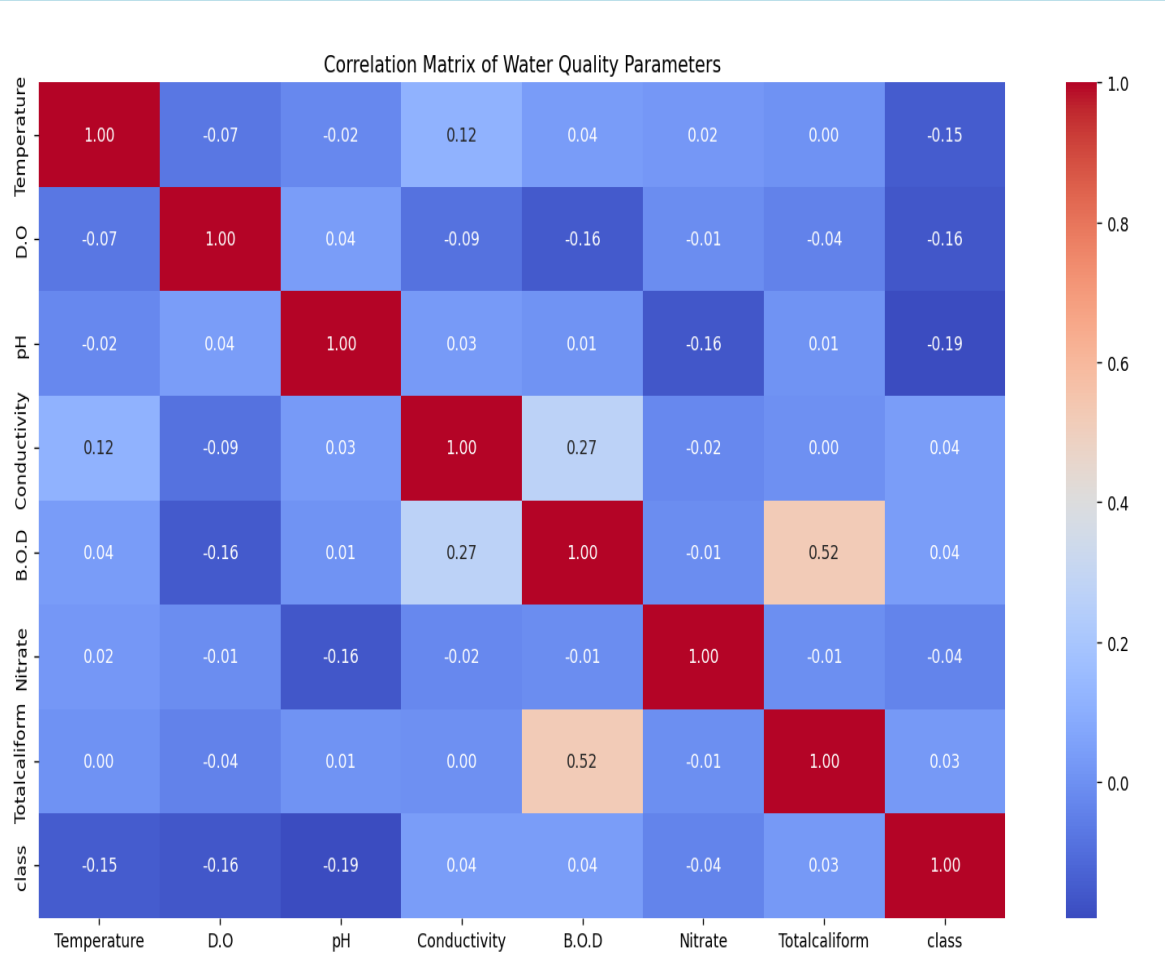
Train dataset description



- Source - Publicly available water quality dataset.
- Samples collected from monitored water bodies across the region shown on the map.
- Total Samples – 1360
- Number of Input Features - 8
- Target Variable - Water quality class (0 = Non-potable, 1 = Potable)

```
Missing values:
Temperature       34
D.O                9
pH                 1
Conductivity      37
B.O.D             68
Nitrate          209
Fecalcaliform    188
Totalcaliform    135
class              0
dtype: int64
```

Train dataset description cont..



Feature Selection & Justification

- Most features show weak-to-moderate correlation, indicating water quality depends on combined effects, not individual parameters.

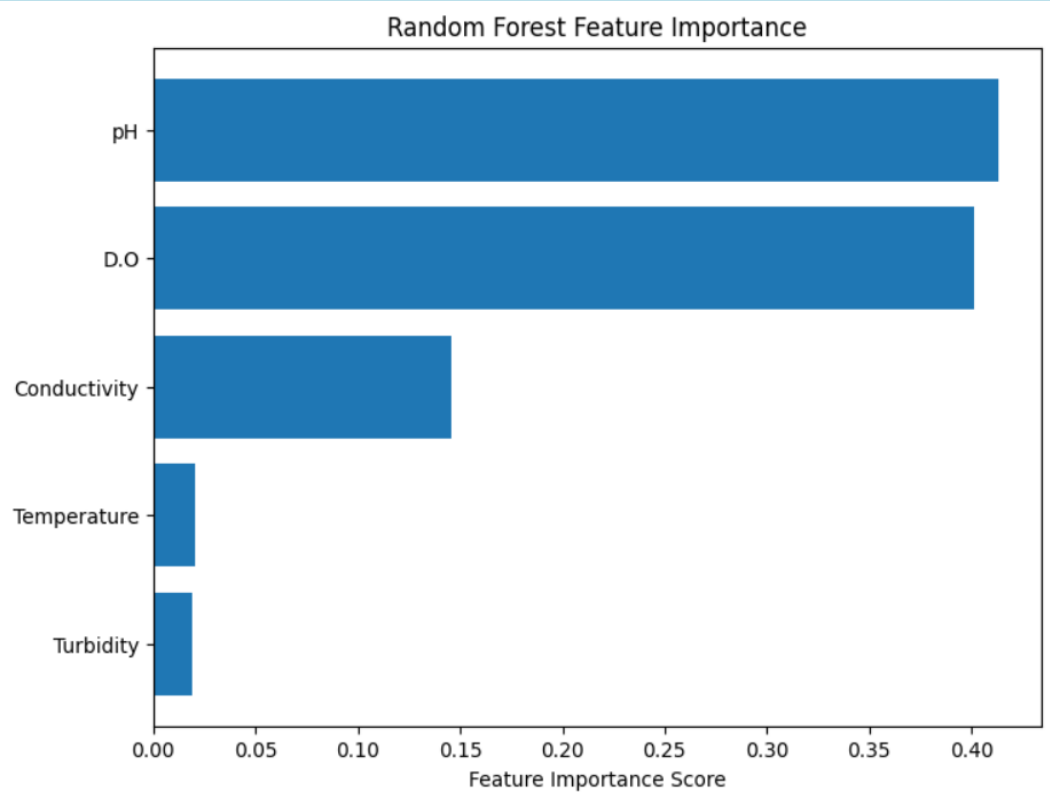
- All relevant physical, chemical, and biological indicators were retained to capture multivariate interactions.

Dropped Features - latitude, longitude, capitalcity, state, fecal coliform

Feature Reduction and Optimal Feature Selection

- Model initially trained using all 8 features
- Performance evaluated with reduced feature subsets
- Accuracy remained high even with fewer parameters
- Applicable highest performance observed with **pH** and **Conductivity**
- These two parameters are common across multiple datasets

Experimental Validation of Combined Features

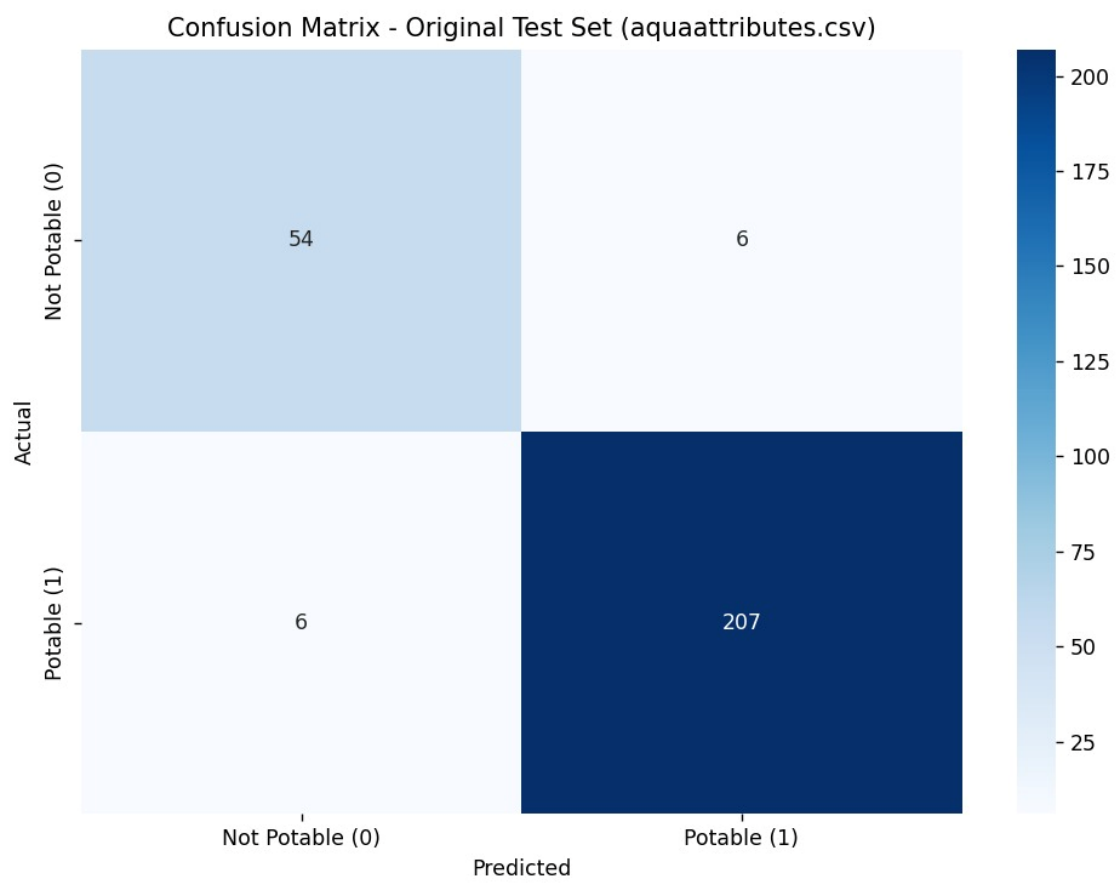


Feature importance diagram

```
◆ Testing subset: ['pH', 'Conductivity']
Accuracy: 0.9560439560439561

◆ Testing subset: ['Temperature', 'B.O.D', 'Nitrate']
Accuracy: 0.7875457875457875

◆ Testing subset: ['pH', 'B.O.D']
Accuracy: 0.9706959706959707

◆ Testing subset: ['pH', 'D.O', 'Nitrate', 'Fecalcaliform']
Accuracy: 0.9743589743589743

◆ Testing subset: ['pH']
Accuracy: 0.978021978021978

◆ Testing subset: ['Fecalcaliform']
Accuracy: 0.7692307692307693

◆ Testing subset: ['D.O']
Accuracy: 0.8315018315018315

◆ Testing subset: ['Totalcaliform']
Accuracy: 0.7655677655677655
```

Accuracy comparison

Results & Performance Analysis



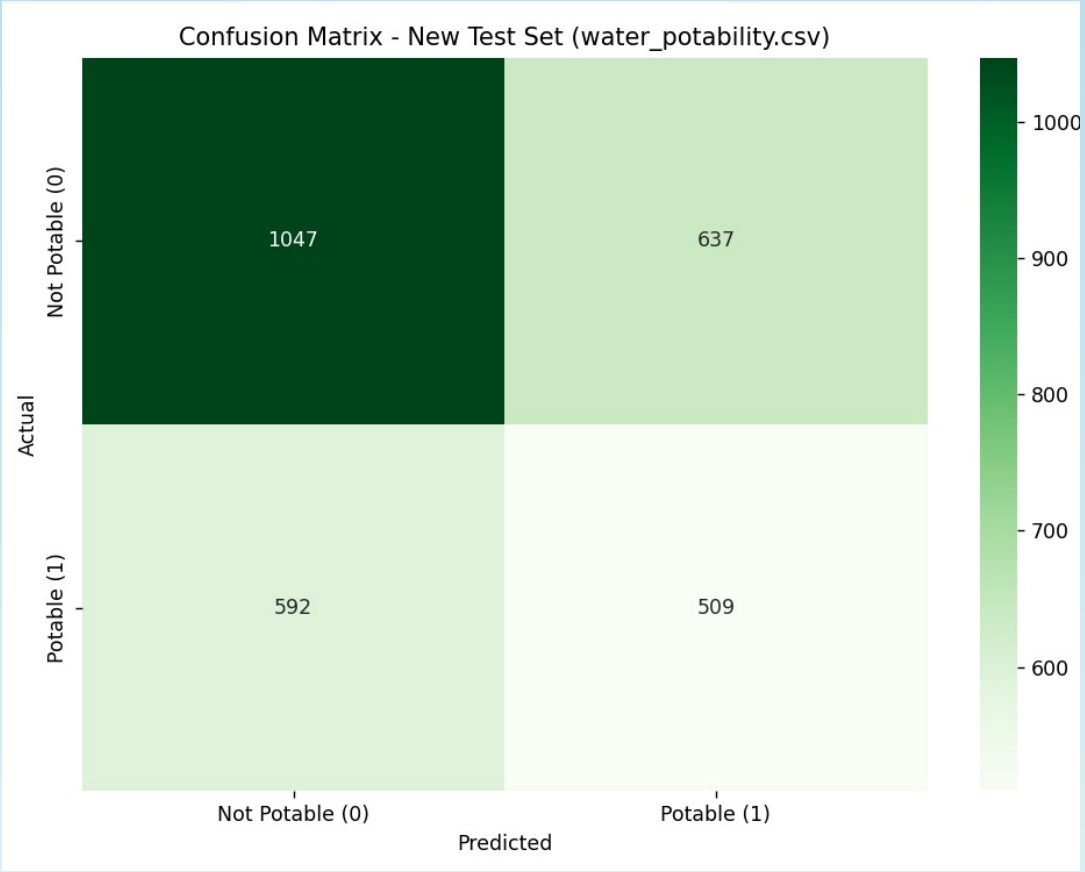

```
✅ Dataset loaded successfully.
Features selected: ['pH', 'Conductivity']
Training Random Forest on pH & conductivity...
Model Accuracy: 0.9560
```

Model Accuracy : 95.60%

Confusion Matrix:

- True Positives: 207
- True Negatives: 54
- False Positives: 6
- False Negatives: 6

Test dataset description




```
         ph  Conductivity  class
0       NaN    564.308654      0
1  3.716080    592.885359      0
2  8.099124    418.606213      0
3  8.316766    363.266516      0
4  9.092223    398.410813      0

🎯 Test Accuracy on water_potability.csv: 0.5587
```

```
Classification Report:
                 precision    recall  f1-score   support

Not Potable (0)       0.64      0.62      0.63      1684
    Potable (1)       0.44      0.46      0.45      1101
```

Test Accuracy: 55.87%
Not-potable:
Precision: 0.64
Recall: 0.62
F1-score: 0.63

Conclusion

- Random Forest successfully captured nonlinear interactions among parameters.
- Model achieved high accuracy (95.60%) on training dataset.
- False Negatives were minimized, reducing safety risk.
- Feature reduction analysis showed pH and Conductivity carry strong predictive power.
- Model demonstrates potential for real-time reservoir monitoring.

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THANK YOU!